

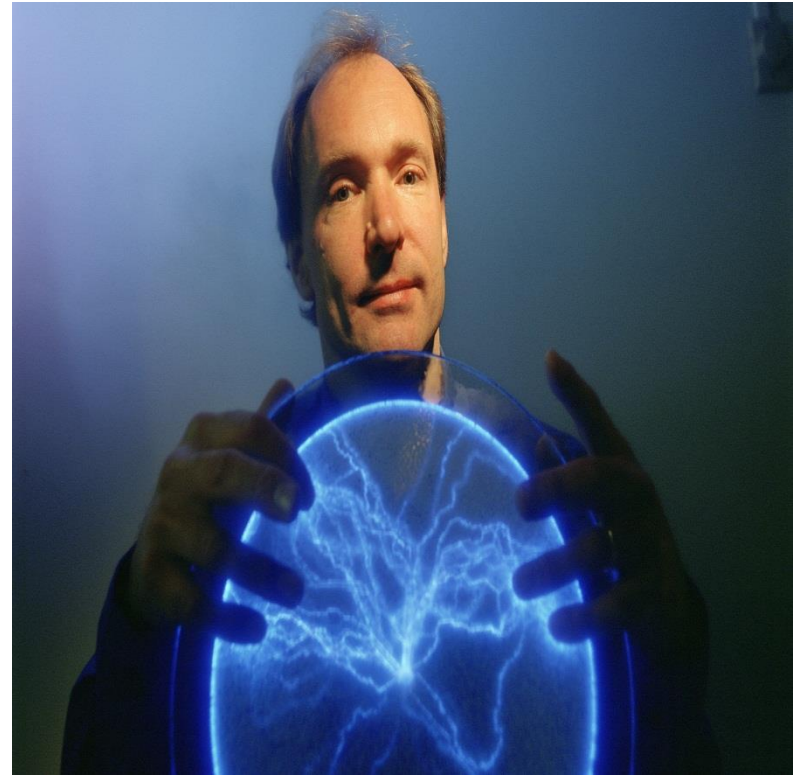
Web Technology

What is World Wide Web?

World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc.



In 1989 the world wide web was invented by Sir Tim Berners-Lee. He was trying to find a new way for scientists to easily share the data from their experiments.



World Wide Web

The WorldWideWeb (W3) is a wide-area [hypermedia](#) information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an [executive summary](#) of the project, [Mailing lists](#) , [Policy](#) , November's [W3 news](#) , [Frequently Asked Questions](#) .

[What's out there?](#)

Pointers to the world's online information, [subjects](#) , [W3 servers](#), etc.

[Help](#)

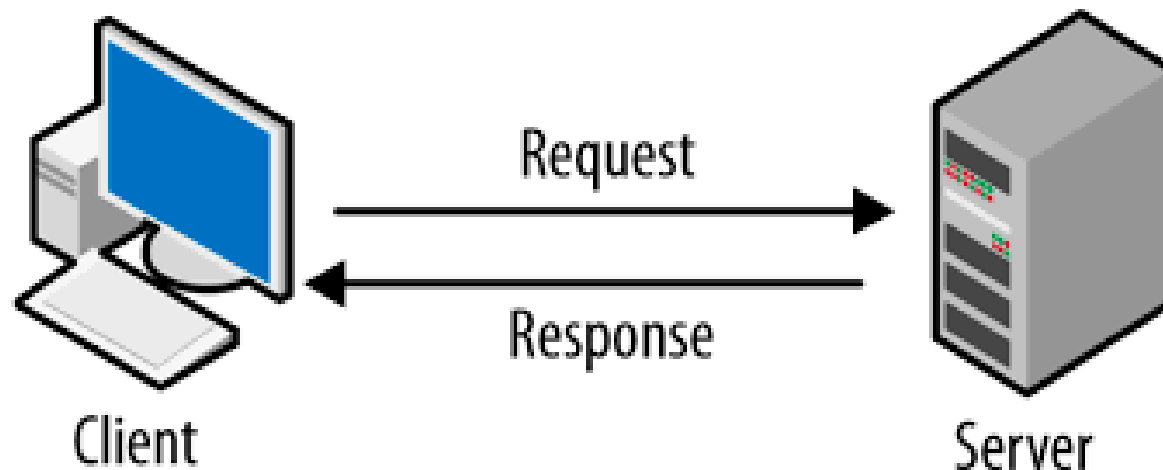
on the browser you are using

[Software Products](#)

A list of W3 project components and their current state. (e.g. [Line Mode](#) ,X11 [Viola](#) , [NeXTStep](#) , [Servers](#) , [Tools](#) , [Mail robot](#) , [Library](#))

A **web browser** (commonly referred to as a **browser**) is application software for accessing the World Wide Web. When a user requests a web page from a particular website, the web browser retrieves the necessary content from a web server and then displays the page on the user's device.

Client Server Architecture is a computing model in which the server hosts, delivers and manages most of the resources and services to be consumed by the client. This type of architecture has one or more client computers connected to a central server over a network or internet connection



The HyperText Markup Language, or HTML is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript.

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. CSS is a cornerstone technology of the World Wide Web

JavaScript. There is no doubt that JavaScript is the King of web development and probably the most popular language among web developers....

HTML

HTML is an acronym which stands for **Hyper Text Markup Language** which is used for creating web pages and web applications.

Hypertext means that the document contains **links that allow the reader to jump to other places** in the document or to another document altogether.

A **Markup Language** is a way that computers speak to each other to control how text is processed and presented. A markup language is used to define the text document within tag which defines the structure of web pages. To do this HTML uses two things: tags and **attributes**.

History of HTML

Tim Berners-Lee is known as the father of HTML.

The first available description of HTML was a document called "HTML Tags" proposed by Tim in late 1991.

HTML Versions

HTML 1.0: The first version of HTML was 1.0, which was the barebones version of HTML language, and it was released in 1991.

HTML 2.0: This was the next version which was released in 1995, and it was standard language version for website design. HTML 2.0 was able to support extra features such as form-based file upload, form elements such as text box, option button, etc.

HTML 3.2: HTML 3.2 version was published by W3C in early 1997. This version was capable of creating tables and providing support for extra options for form elements. It can also support a web page with complex mathematical equations. It became an official standard for any browser till January 1997. Today it is practically supported by most of the browsers.

HTML Versions

HTML 4.01: HTML 4.01 version was released on December 1999, and it is a very stable version of HTML language. This version is the current official standard, and it provides added support for stylesheets (CSS) and scripting ability for various multimedia elements.

HTML5: HTML5 is the newest version of HyperText Markup language. The first draft of this version was announced in January 2008. There are two major organizations one is W3C (World Wide Web Consortium), and another one is WHATWG(Web Hypertext Application Technology Working Group) which are involved in the development of HTML5 version, and still, it is under development.

Note: HTML is a case-insensitive language, which means we can use tags either in lower-case or upper-case

Building blocks of HTML

HTML uses predefined tags and elements which tell the browser how to properly display the content. Remember to include closing tags.

Tags: HTML is a tag based language. They are defined within the angle brackets (< and >)



Attribute: An attribute in HTML provides extra information about the element, and it is applied within the start tag. An HTML attribute contains two fields: name & value.

Syntax

<**tag** name attribute_name= "attr_value"> content </ **tag** name>

Elements: An HTML element is an individual component of an HTML file. In an HTML file, everything written within tags are termed as HTML elements.

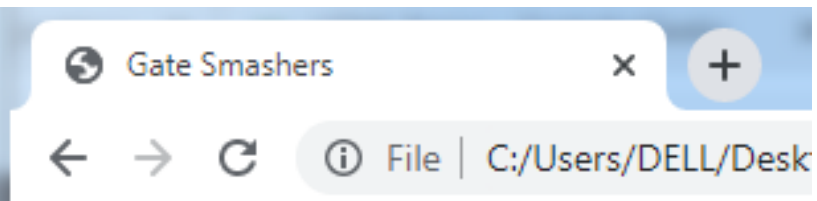
<p align="left"> Left aligned paragraph </p>

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first.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Gate Smashers</title>
5 </head>
6 <body>
7
8 <h2 style="color:Tomato;">
  Welcome Smashers</h2>
9
10 </body>
11 </html>
```



Welcome Smashers

HTML page structure

The basic structure of an HTML page is laid out below. It contains the essential building-block elements (i.e. doctype declaration, HTML, head, title, and body elements) upon which all web pages are created.

`<!DOCTYPE html>` (information about HTML version)

`<html>` (Root element)

`<head>` (contains HTML metadata)

`<title> Title of Webpage </title>` ← Title of HTML page

`</head>`

`<body>` (contains content of HTML)

`<h2> Heading </h2>` ← Heading tag

`<p> Paragraph </p>` ← Paragraph tag

`</body>`

`</html>`

<DOCTYPE! html>: This is the document type declaration (not technically a tag). It declares a document as being an HTML document. The doctype declaration is not case-sensitive.

<html>: This is called the HTML root element. All other elements are contained within it.

<head>: The head tag contains the “behind the scenes” elements for a webpage. Elements within the head aren’t visible on the front-end of a webpage. HTML elements used inside the <head> element include:

<style>

<title>

<base>

<noscript>

<script>

<meta>

<link>

<body>: The body tag is used to enclose all the visible content of a webpage. In other words, the body content is what the browser will show on the front-end.

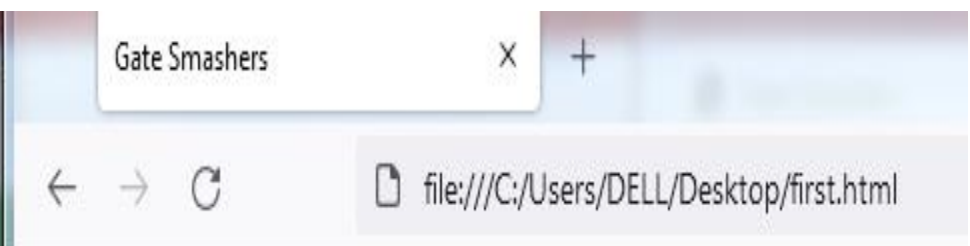
An HTML document can be created using any text editor. Save the text file using **.html** or **.htm**. Once saved as an HTML document, the file can be opened as a webpage in the browser.

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first.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Gate Smashers</title>
5 </head>
6 <body>
7
8 <h1>Heading tag</h1>
9
10 </body>
11 </html>
```



Heading tag

HTML Text Editors

These are used to create and modify web pages. HTML codes can be written in any text editors including the **Notepad**. One just needs to write HTML in any text editor and save the file with an extension “.html”. Some of the popular HTML text editors are given below:

- Notepad
- Notepad++
- Sublime Text 3
- Atom

Comments in HTML

The Comment tag is used to insert comments in the source code. Comments are **not displayed in the browsers**.

You can use comments to explain your code, which can help you when you edit the source code at a later date. This is especially useful if you have a lot of code.

Syntax:

<!--comment-->

DOCTYPE

The **DOCTYPE** declaration describes **what version of HTML the page is written in**. It is the very first thing in your HTML document that you see in every web page. It is written at the top of every page before the `<html>` tag.

The doctype declaration is not an HTML tag. It is the one recommended by HTML5.

<!DOCTYPE> declaration is NOT case sensitive

Syntax for doctype is: **<!DOCTYPE html>**.

Paragraphs

Paragraphs are **blocks of text** separated from each other by some space. They are defined using the **<p>** and **</p>** tags. When p element ends, the next element appears in next line.

```
<!DOCTYPE html>
<html>
<head>
<title>p tag</title>
</head>
<body>
<p>This is paragraph1.</p>
<p>This is paragraph2.</p>
<!-- without using p-tag -->
Without paragraph line 1.
Without paragraph line 2.
Without paragraph line 3.
</body>
</html>
```

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```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>p tag</title>
5 </head>
6 <body>
7 <p>This is paragraph1.</p>
8 <p>This is paragraph2.</p>
9 <!-- ithout using p-tag -->
10 Without paragraph line 1.
11 Without paragraph line 2.
12 Without paragraph line 3.
13 </body>
14 </html>
```

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This is paragraph1.

This is paragraph2.

Without paragraph line 1. Without paragraph line 2. Without paragraph line 3.

Headings Tag

These are tags in HTML to mark some content as headings. In fact, there are six different levels of headings h1, h2, h3, h4, h5 and h6. Among them h1 defines the largest heading and h6 defines the smallest level heading.

```
<!DOCTYPE html>
<html>
<head>
<title>Heading Tag</title>
</head>
<body>
<h1>Gate Smashers</h1>
<h2>Gate Smashers</h2>
<h3>Gate Smashers</h3>
<h4>Gate Smashers</h4>
<h5>Gate Smashers</h5>
<h6>Gate Smashers</h6>
</body>
</html>
```

```
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first.html
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Heading Tag</title>
5 </head>
6 <body>
7 <h1>Gate Smashers</h1>
8 <h2>Gate Smashers</h2>
9 <h3>Gate Smashers</h3>
10 <h4>Gate Smashers</h4>
11 <h5>Gate Smashers</h5>
12 <h6>Gate Smashers</h6>
13 </body>
14 </html>
15
```



BREAK Tag

`<br>` tag is used to introduce a **single line break** between the contents. This means that when this tag is used in between a single line, the content after this tag will move to the next line. Do not use it to provide space between block of elements (eg., paragraph and heading).

Eg:

```
<h3>Welcome to<br>Gate Smashers</h3>
```

HTML Formatting

HTML provides us ability to format text without using CSS. There are many formatting tags in HTML. These tags are used to make text bold, italicized, or underlined.

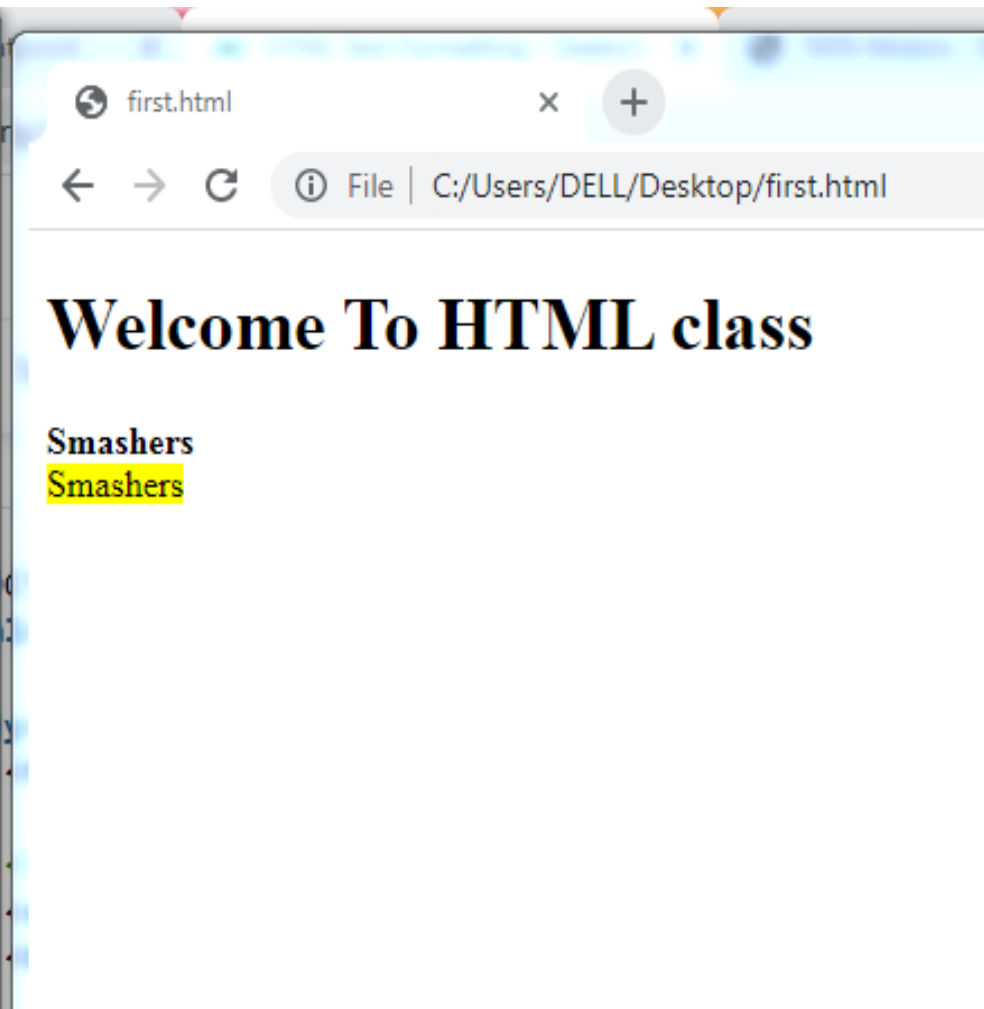
```
<!DOCTYPE html>
<html>
  <body>
    <h2>Welcome To HTML class</h2>

    <!--Text in Bold-->
    <strong>Smashers</strong>
    <br>

    <!--Text in Highlight-->
    <mark>Smashers</mark>
  </body>
</html>
```

```
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first.html
1 <!DOCTYPE html>
2 <html>
3
4 <body>
5   <h1>Welcome To HTML class</h1>
6
7   <!--Text in bold-->
8   <b>Smashers</b>
9   <br>
10
11   <!--Text in Highlight-->
12   <mark>Smashers</mark>
13 </body>
14
15 </html>
16
```



HTML Lists

HTML Lists are used to specify information in the form of lists . All lists may contain one or more list elements. There are three types of HTML lists:

- i) Ordered List or Numbered List (ol)
- ii) Unordered List or Bulleted List (ul)
- iii) Description List or Definition List (dl)

We can create a list inside another list, which will be termed as nested List.

HTML Ordered List

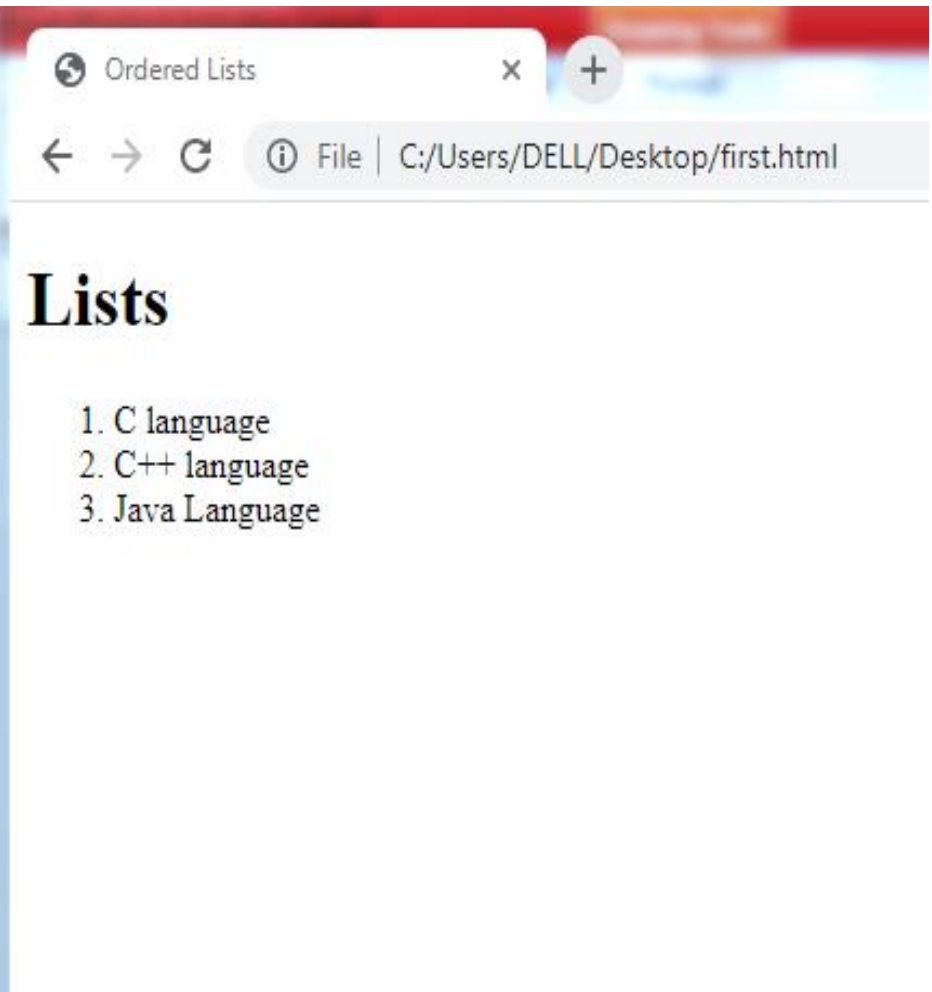
In the ordered HTML lists, all the list items are marked with numbers by default. It is known as numbered list also. The ordered list starts with `<ol>` tag and the list items start with `<li>` tag. They are defined using the `<ol>` tag.

Eg:

```
<!DOCTYPE html>
<html>
<head>
<title>Ordered Lists</title>
</head>
<body>
<h1>Lists </h1>
<ol>
<li>C language</li>
<li>C++ language</li>
<li>Java Language</li>
</ol>
</body>
</html>
```

```
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first.html
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Ordered Lists</title>
5 </head>
6 <body>
7 <h1>Lists </h1>
8 <ol>
9 <li>C language</li>
10 <li>C++ language</li>
11 <li>Java Language</li>
12 </ol>
13 </body>
14 </html>
15
```



HTML Unordered List

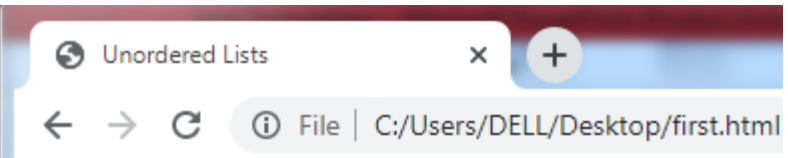
Unordered List is used to group a set of items with no particular order. By default the items are followed by bullets. They are defined using the **** tag.

Eg:

```
<!DOCTYPE html>
<html>
<head>
<title>Unordered Lists</title>
</head>
<body>
<h1>Lists</h1>
<ul>
<li>Data structure</li>
<li>C language</li>
<li>C++ Language</li>
</ul>
</body>
</html>
```

```
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first.html
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Unordered Lists</title>
5 </head>
6 <body>
7 <h1>Lists </h1>
8 <ul>
9 <li>C language</li>
10 <li>C++ language</li>
11 <li>Java Language</li>
12 </ul>
13 </body>
14 </html>
15
```



Lists

- C language
- C++ language
- Java Language

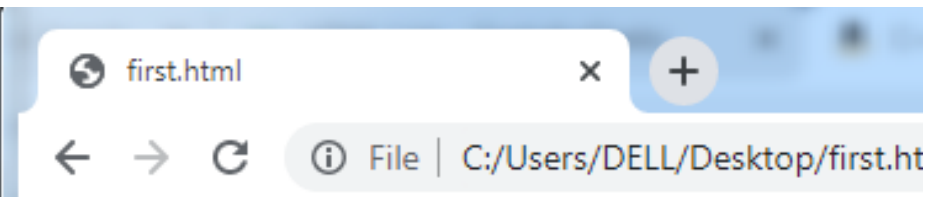
HTML Description List

A description list is a list of terms, with a description of each term. A definition list starts with the `<dl>` tag. Each definition - list term starts with the `<dt>` tag. Each definition - list definition starts with the `<dd>` tag.

```
<!DOCTYPE html>
<html>
<body>
  <h2>Description List</h2>
  <dl>
    <dt>C Programming Language</dt>
    <dd>- By Dennis Ritchie</dd>
    <dt>C++ Programming Language</dt>
    <dd>- By Bjarne Stroustrup</dd>
  </dl>
</body>
</html>
```

```
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first.html
1 <!DOCTYPE html>
2 <html>
3
4 <body>
5   <h2>Description List</h2>
6   <dl>
7     <dt>C Programming Language</
      dt>
8     <dd>- By Dennis Ritchie</dd>
9     <dt>C++ Programming Language
      </dt>
10    <dd>- By Bjarne Stroustrup</
      dd>
11    </dl>
12  </body>
13
14 </html>
```



Description List

C Programming Language

- By Dennis Ritchie

C++ Programming Language

- By Bjarne Stroustrup

HTML Tables

HTML Table is an arrangement of data in rows and columns. Tables are widely used in communication, research, and data analysis. It is used to compare two or more items in the tabular form data.

HTML table is created with the “table” tag. Each table row is created with the “tr” tag. A table header is created with the “th” tag. By default, table headings are bold and centered. A table data is defined with the “td” tag.

```
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first.html
1 <!DOCTYPE html>
2 <html>
3
4 <body>
5   <table>
6     <tr>
7       <th>EName</th>
8       <th>EId</th>
9       <th>Department</th>
10    </tr>
11    <tr>
12      <td>Naina</td>
13      <td>2438</td>
14      <td>Computer Science</td>
15    </tr>
16    <tr>
17      <td>varun</td>
18      <td>3456</td>
19      <td>computer Science</td>
20    </tr>
21    <tr>
22      <td>Ananya</td>
23      <td> 8754</td>
24      <td>Electronics and
25        Communications</td>
26    </tr>
27  </table>
28 </body>
```

first.html

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EName	EId	Department
Naina	2438	Computer Science
varun	3456	computer Science
Ananya	8754	Electronics and Communications

```
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first.html
1 <!DOCTYPE html>
2 <html>
3
4 <body>
5 <table border="1">
6   <tr>
7     <th>EName</th>
8     <th>EId</th>
9     <th>Department</th>
10  </tr>
11  <tr>
12    <td>Naina</td>
13    <td>2438</td>
14    <td>Computer Science</td>
15  </tr>
16  <tr>
17    <td>varun</td>
18    <td>3456</td>
19    <td>computer Science</td>
20  </tr>
21  <tr>
22    <td>Ananya</td>
23    <td> 8754</td>
24    <td>Electronics and
    Communications</td>
25  </tr>
26 </table>
27 </body>
```

first.html

File | C:/Users/DELL/Desktop/first.html

EName	EId	Department
Naina	2438	Computer Science
varun	3456	computer Science
Ananya	8754	Electronics and Communications

HTML Anchor

The **HTML anchor tag** defines *a hyperlink that links one page to another page*. It can create hyperlink to other web page as well as files, location, or any URL. The "href" attribute is the most important attribute of the HTML a tag. and which links to destination page or URL.

href attribute of HTML anchor tag

The href attribute is used to define the address of the file to be linked. In other words, it points out the destination page. The syntax of HTML anchor tag is given below:

`<a href = "....."> Link Text </a>`

`<a href="first.html">Click Here</a>`

Target attribute of HTML anchor tag

HTML Image

HTML img tag is used to display image on the web page. HTML img tag is an empty tag that contains attributes only, closing tags are not used in HTML image element.

```
<h1>HTML Image Example</h1>
```

```

```

height and width attribute with img tag

You have learnt about how to insert an image in your web page, now if we want to give some height and width to display image according to our requirement, then we can set it with height and width attributes of image.

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first.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Image tag</title>
5 </head>
6 <body>
7 <h1>HTML image example</h1>
8 
10 </html>
```

image tag

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HTML image example




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first.html x

1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Image tag</title>
5 </head>
6 <body>
7 <h1>HTML image example</h1>
8 
</html>
```



```
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first.html x
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Image tag</title>
5 </head>
6 <body>
7 <h1>HTML image example</h1>
8 
10 </html>
```

